THE LOCAL DIMENSION AT A POINT IS UNCHANGED BY EXTENDING THE BASE FIELD

THE STACKS PROJECT AUTHORS

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Allowed standard prerequisites (editorial).

- Presentation of the base change by extending the ideal. Used at the start of the counted proof (lines 28537-28540). Describes the extended algebra as a quotient of a polynomial ring by the extended ideal.
- The dimension formula for a point of a finite type algebra over a field. Used at the end of the counted proof (lines 28560-28571). Expresses both local dimensions through the same numerical data, which is what makes them equal.

Source reference. Original lemma 00P2: sources/stacks-algebra/pinned-original-source.tex, source lines 28487–28487. Navigation pointer for \ref{lemma-codimension} used inside the counted span (line 28563); the argument that consumes it is retained in the span.

Source reference. Original lemma 00ON: sources/stacks-algebra/pinned-original-source.tex, source lines 27638–27638. Navigation pointer for \ref{lemma-dimension-base-fibre-equals-total} used inside the counted span (line 28557); the argument that consumes it is retained in the span.

Date: stacks-project:algebra.tex:local-snapshot-20261012; GNU FDL 1.2 or later.

Lemma 0.1. *Let k be a field. Let S be a finite type k-algebra. Set $X = \mathrm{Spec}(S)$. Let K/k be a field extension. Set $S_K = K \otimes_k S$, and $X_K = \mathrm{Spec}(S_K)$. Let $\mathfrak{q} \subset S$ be a prime corresponding to $x \in X$ and let $\mathfrak{q}_K \subset S_K$ be a prime corresponding to $x_K \in X_K$ lying over $\mathfrak{q}$. Then $\dim_x X = \dim_{x_K} X_K$.*

Proof. Choose a presentation $S = k[x_1, \ldots, x_n]/I$. This gives a presentation $K \otimes_k S = K[x_1, \ldots, x_n]/(K \otimes_k I)$. Let $\mathfrak{q}'_K \subset K[x_1, \ldots, x_n]$, resp. $\mathfrak{q}' \subset k[x_1, \ldots, x_n]$ be the corresponding primes. Consider the following commutative diagram of Noetherian local rings

$$\begin{array}{ccc} K[x_1, \ldots, x_n]_{\mathfrak{q}'_K} & \longrightarrow & (K \otimes_k S)_{\mathfrak{q}_K} \\ \uparrow & & \uparrow \\ k[x_1, \ldots, x_n]_{\mathfrak{q}'} & \longrightarrow & S_{\mathfrak{q}} \end{array}$$

Both vertical arrows are flat because they are localizations of the flat ring maps $S \to S_K$ and $k[x_1, \ldots, x_n] \to K[x_1, \ldots, x_n]$. Moreover, the vertical arrows have the same fibre rings. Hence, we see from Lemma 00ON that $\mathrm{height}(\mathfrak{q}') - \mathrm{height}(\mathfrak{q}) = \mathrm{height}(\mathfrak{q}'_K) - \mathrm{height}(\mathfrak{q}_K)$. Denote $x' \in X' = \mathrm{Spec}(k[x_1, \ldots, x_n])$ and $x'_K \in X'_K = \mathrm{Spec}(K[x_1, \ldots, x_n])$ the points corresponding to $\mathfrak{q}'$ and $\mathfrak{q}'_K$. By Lemma 00P2 and what we showed above we have

$$\begin{aligned} n - \dim_x X &= \dim_{x'} X' - \dim_x X \\ &= \mathrm{height}(\mathfrak{q}') - \mathrm{height}(\mathfrak{q}) \\ &= \mathrm{height}(\mathfrak{q}'_K) - \mathrm{height}(\mathfrak{q}_K) \\ &= \dim_{x'_K} X'_K - \dim_{x_K} X_K \\ &= n - \dim_{x_K} X_K \end{aligned}$$

and the lemma follows. □